

DSPC 7514 Deliverable 2: Project Proposal (SDG Analysis)

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1 Load necessary libraries

```
library(tidyverse)
library(readxl)
library(devtools)
library(vdemdata) # call vdem package
library(ERT) # call ERT package
library(WDI) # call WDI package for GINI coefficient
library(janitor)
library(knitr)
```

1.1 Loading SDG Data

Using most recent SDG data from the Sustainable Development Report 2025

```
# Filter-out pseudo countries:
ids_to_drop <- c(
  "_Africa", "_BRICS", "_BRICSplus", "_E_Euro_Asia", "_E_S_Asia", "_HIC",
  "_LAC", "_LIC", "_LIC_LMIC", "_LMIC", "_MENA", "_OECD", "_Oceania",
  "_SIDS", "_UMIC", "_World", "_LAC"
)

# sdg excel from directory (Raw indicators)
sdg_raw <- read_excel("data/input/SDR2025-data.xlsx", sheet = "All Raw Data") %>%
  filter(year >= 2000) %>%
  filter(!id %in% ids_to_drop)
```

1.2 SDG Data Prep (reshaping from wide to long format)

```
sdg_long <- sdg_raw %>%
  pivot_longer(
    cols = starts_with("sdg"),
    names_to = "code",
    values_to = "value"
```

```
) %>%
mutate(value = suppressWarnings(as.numeric(value))) # in case of text like ".."
```

2 1) Creating country–year missingness variable

Time period: 2000–2025 (SDG era: 2015–2025)

```
# CREATING MISSINGNESS VARIABLE
sdg_missing_cy <- sdg_long %>%
  group_by(Country, year) %>%
  summarise(
    total_indicators = n(),
    missing_indicators = sum(is.na(value)),
    missing_rate = missing_indicators / total_indicators, # VARIABLE OF INTEREST
    .groups = "drop"
  )

# SUMMARIZING MISSINGNESS VARIABLE
sdg_summary <- sdg_missing_cy %>%
  summarise(
    mean_missing = mean(missing_rate, na.rm = TRUE),
    sd_missing = sd(missing_rate, na.rm = TRUE),
    min_missing = min(missing_rate, na.rm = TRUE),
    max_missing = max(missing_rate, na.rm = TRUE),
    n_countries = n_distinct(Country),
    n_obs = n()
  )
sdg_summary
```

```
## # A tibble: 1 x 6
##   mean_missing sd_missing min_missing max_missing n_countries n_obs
##   <dbl>        <dbl>        <dbl>        <dbl>        <int> <int>
## 1      0.562      0.147      0.198      0.968         193  5018
```

```
kable(sdg_summary, caption = "Summary Statistics: SDG Data Missingness (2000–2025), country-years")
```

Table 1: Summary Statistics: SDG Data Missingness (2000–2025), country-years

mean_missing	sd_missing	min_missing	max_missing	n_countries	n_obs
0.5616954	0.1467174	0.1984127	0.968254	193	5018

Interpretation On average, 56.2% of SDG indicators are missing for a given country in a given year — meaning countries, on average, report only about 43.8 of SDG indicators.

The best-reported country-year had only 19.8 % missing, while the worst had 96.8%.

There's moderate variation across countries and years — some countries are much more complete than others. The data has a standard deviation of 14.7 percentage points.

The data includes 193 countries or territories, consistent with global SDG coverage.

3 2) Missigness (Variation) Across Time

```
# summarizing missingness by year
sdg_missing_y <- sdg_missing_cy %>%
  group_by(year) %>%
  summarise(
    mean_missing = mean(missing_rate, na.rm = TRUE),
    sd_missing = sd(missing_rate, na.rm = TRUE),
    median_missing = median(missing_rate, na.rm = TRUE),
    n_countries = n_distinct(Country))

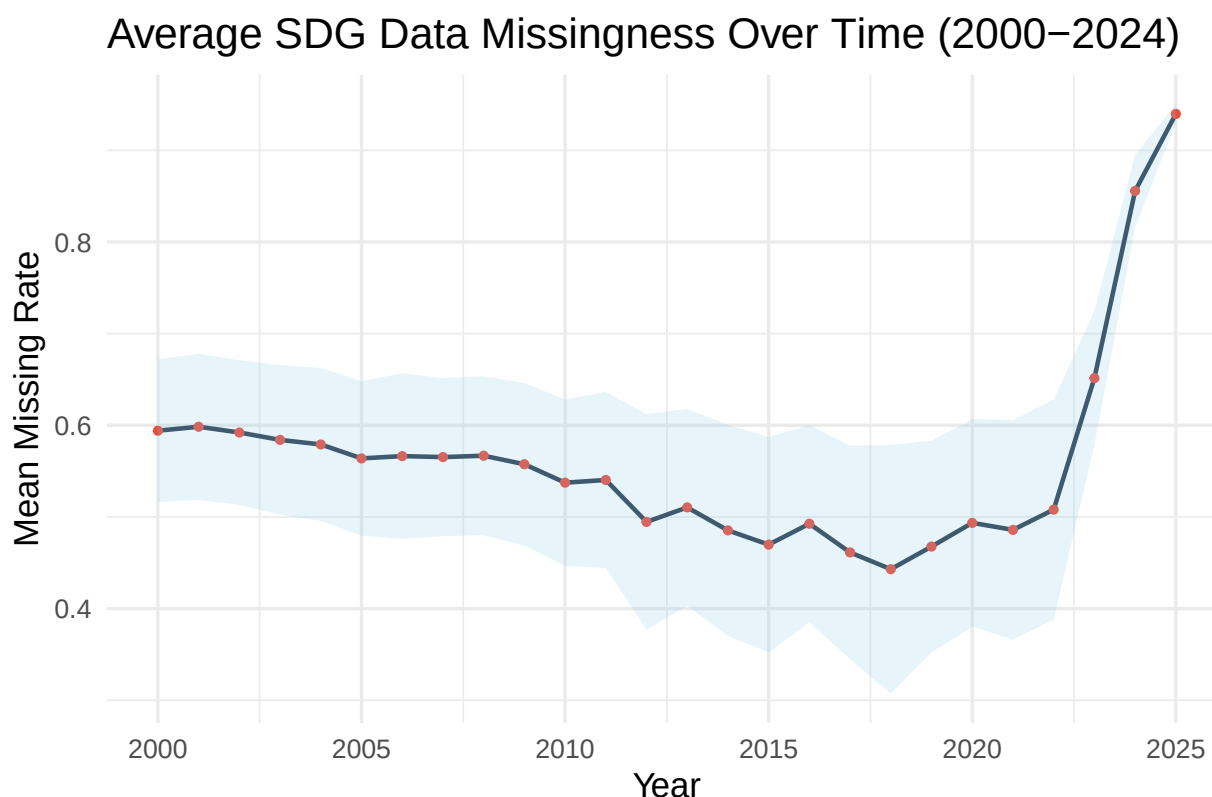
kable(sdg_missing_y, caption = "Summary Statistics: SDG Data Missingness (2000-2025), yearly stats")
```

Table 2: Summary Statistics: SDG Data Missingness (2000–2025), yearly stats

year	mean_missing	sd_missing	median_missing	n_countries
2000	0.5940867	0.0778500	0.5793651	193
2001	0.5984045	0.0796915	0.5793651	193
2002	0.5921128	0.0789051	0.5714286	193
2003	0.5840530	0.0816243	0.5634921	193
2004	0.5790772	0.0833781	0.5634921	193
2005	0.5638622	0.0842623	0.5396825	193
2006	0.5663706	0.0903860	0.5555556	193
2007	0.5653014	0.0862810	0.5476190	193
2008	0.5667818	0.0868057	0.5555556	193
2009	0.5574883	0.0886062	0.5476190	193
2010	0.5373797	0.0907575	0.5238095	193
2011	0.5403405	0.0960897	0.5317460	193
2012	0.4945308	0.1175101	0.4841270	193
2013	0.5104038	0.1074171	0.5000000	193
2014	0.4853195	0.1151800	0.4682540	193
2015	0.4697755	0.1174578	0.4682540	193
2016	0.4925981	0.1076196	0.4841270	193
2017	0.4612633	0.1163611	0.4523810	193
2018	0.4429641	0.1356932	0.4365079	193
2019	0.4676783	0.1154792	0.4603175	193
2020	0.4935439	0.1132334	0.4920635	193
2021	0.4858541	0.1198809	0.4761905	193
2022	0.5079776	0.1199823	0.5079365	193
2023	0.6514927	0.0729453	0.6349206	193
2024	0.8557036	0.0394032	0.8492063	193
2025	0.9397154	0.0105065	0.9365079	193

```
# Missingness over time line graph
ggplot(sdg_missing_y, aes(x = year, y = mean_missing)) +
  geom_line(color = "#2C3E50", size = 0.8) +
  geom_point(color = "#E74C3C", size = 1) +
  geom_ribbon(aes(ymin = mean_missing - sd_missing, ymax = mean_missing + sd_missing),
    fill = "skyblue", alpha = 0.2) +
  theme_minimal(base_size = 13) +
```

```
labs(
  title = "Average SDG Data Missingness Over Time (2000-2024)",
  x = "Year",
  y = "Mean Missing Rate",
  caption = "Source: Sustainable Development Report (2025)")
```



Source: Sustainable Development Report (2025)

Interpretation The Average SDG Data Missingness Over Time graph shows the trend in the global SDG data missingness rate from 2000 to 2025. Between 2000 and 2019, the global average missingness steadily declined from roughly 60% to below 45% in 2019, reflecting improvements in data coverage and reporting consistency under the early SDG framework.

However, after 2019, the trend reverses sharply. By 2025, the global mean missingness reaches 94%, indicating that only about 6 % of SDG indicators are being reported on average.

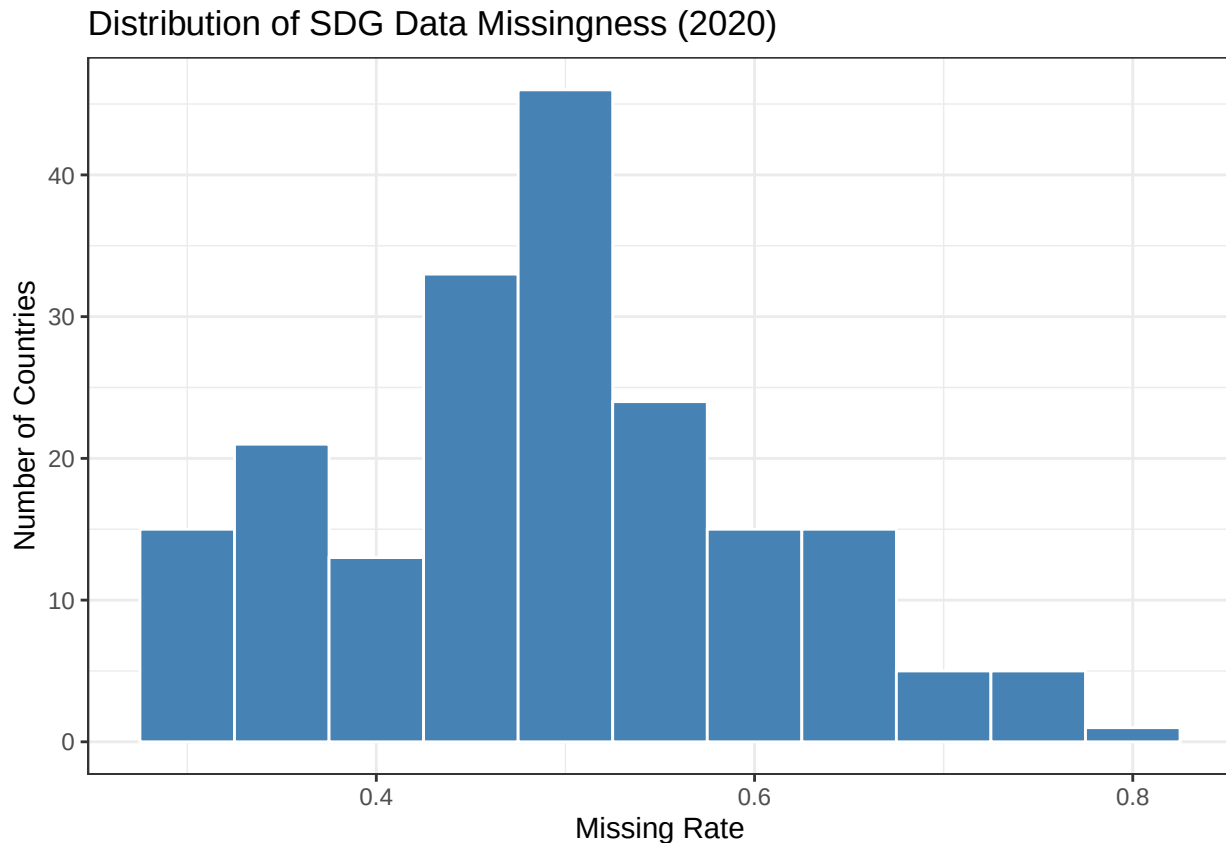
The relative increase in missingness between 2019 and 2021 is particularly striking, with the mean missingness rate jumping from around 46.8% to 48.6%, suggesting a significant disruption in SDG data reporting, which coincides with post-pandemic disruptions, capacity shortfalls, and possible political restrictions on data transparency.

Across the entire 2015–2024 period, countries report an average of 43.8% of SDG indicators, with a standard deviation of 14.7 percentage points based on 5,018 country-year observations covering 193 countries. The lowest level is around 44.3% in 2018 and the highest reaching 94% in 2025.

Overall, the figure suggests that initial progress toward comprehensive SDG reporting has been followed by a dramatic reversal, which may not necessarily reflect a decline in reporting but a lag in data collection and processing. This overall trend cannot alone provide evidence to suggest a systemic decline in reporting capacity because it can take time for data to be collected, processed, and published, especially in the wake of global disruptions.

4 3) Cross-Sectional Distribution (2020 snapshot)

```
sdg_missing_cy %>%  
  filter(year == 2020) %>%  
  ggplot(aes(x = missing_rate)) +  
  geom_histogram(binwidth = 0.05, fill = "steelblue", color = "white") +  
  theme_bw() +  
  labs(title = "Distribution of SDG Data Missingness (2020)",  
        x = "Missing Rate", y = "Number of Countries")
```



Interpretation The Distribution of SDG Data Missingness graph shows the distribution of SDG data missingness across countries in 2020. The average missingness rate that year was 49.4 %, with a standard deviation of 11.3 percentage points.

Most countries reported between 31% and 77.8% of their SDG indicators, indicating moderate variation in reporting completeness across countries.

The histogram shows a concentration around the mid-range (roughly 40–50%), suggesting that while many countries achieved partial reporting, full SDG data coverage remained uncommon in 2020.

5 4) Within-Country Variation

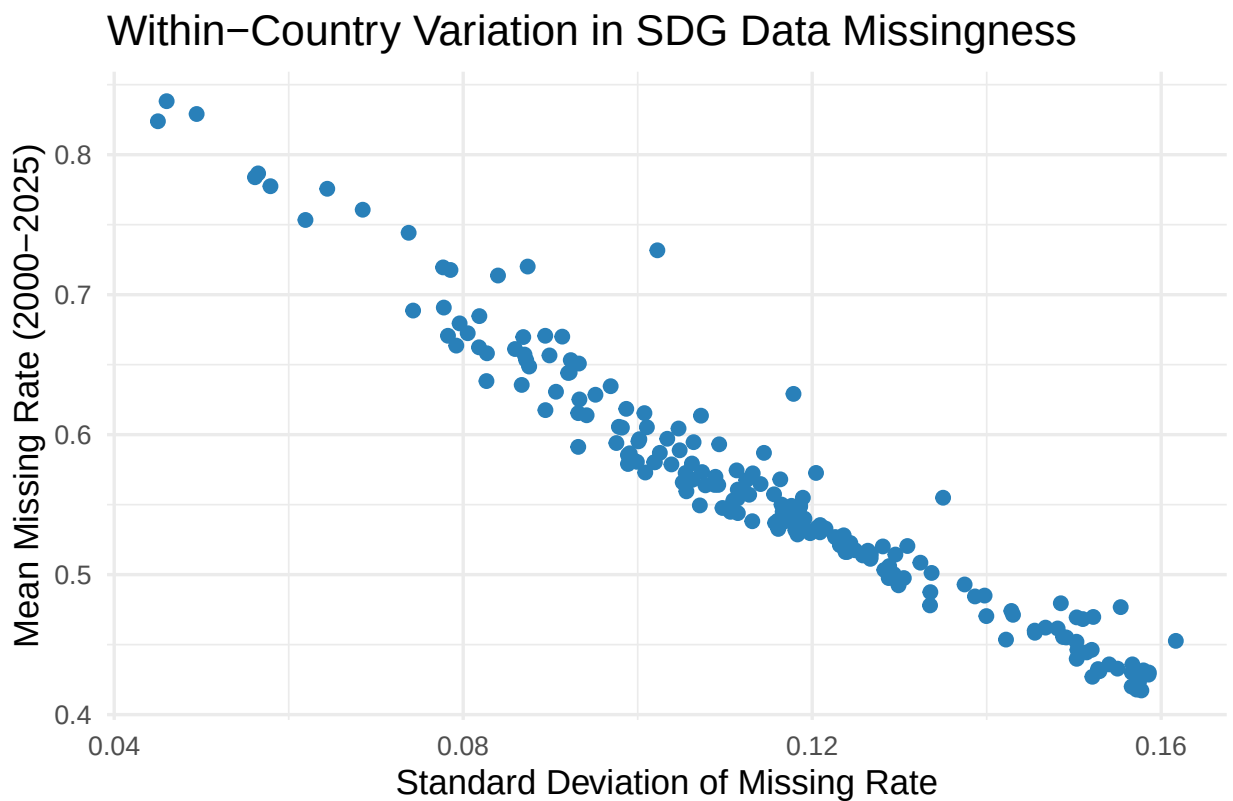
```
sdg_within <- sdg_missing_cy %>%  
  group_by(Country) %>%
```

```

summarise(
  mean_missing = mean(missing_rate, na.rm = TRUE),
  sd_missing = sd(missing_rate, na.rm = TRUE),
  n_years = n()
) %>%
  arrange(desc(mean_missing))

# scatterplot of mean missingness vs. within-country variation
ggplot(sdg_within, aes(x = sd_missing, y = mean_missing)) +
  geom_point(color = "#2980B9", size = 2) +
  theme_minimal(base_size = 13) +
  labs(
    title = "Within-Country Variation in SDG Data Missingness",
    x = "Standard Deviation of Missing Rate",
    y = "Mean Missing Rate (2000-2025)",
    caption = "Source: Sustainable Development Report (2025)"
  )

```



Source: Sustainable Development Report (2025)

6 Ensuring pseudo countries are filtered across datasets